

Lecture - 10

- $X_1, X_2, \dots, X_n$, $E[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$
 $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, $E[\bar{X}] = \mu$, $\text{Var}(\bar{X}) = \sigma^2/n$

$SD = SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$ $X_i \sim N(\mu, \sigma^2)$
 $\bar{X} \sim N(\mu, \sigma^2/n)$

$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$

- CI for μ when σ is known

$P(-Z_{\alpha/2} < Z < Z_{\alpha/2}) = 1 - \alpha$
 $\Rightarrow P(-Z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < Z_{\alpha/2}) = 1 - \alpha$

$\Rightarrow P(-Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}) = 1 - \alpha$

$\therefore$, $100(1-\alpha)\%$ CI is

$\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

or $\boxed{\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}}$
 (1.96)

95% probability that μ is inside the CI.

- for 95%, $\alpha = 0.05$, $\alpha/2 = 0.025$ $Z_{0.025} = 1.96$
 for 99%, $\alpha = 0.01$, $\alpha/2 = 0.005$ $Z_{0.005} = 2.575$
 for 90%, $\alpha = 0.1$, $\alpha/2 = 0.05$ $Z_{0.05} = 1.645$
 higher confidence $\rightarrow$ wider interval

Q. $n = 36$, $\bar{X} = 2.6$, $\sigma = 0.3$

95% $\Rightarrow Z_{\alpha/2} = 1.96 \Rightarrow \cancel{2.6 \pm 0.3} \Rightarrow 2.6 \pm 0.05$
 $\cancel{2.6 \times 10}$

$\cancel{(2.55, 2.65)}$
 $\cancel{99\%} \Rightarrow \cancel{Z_{\alpha/2} = 2.575} \Rightarrow 2.6 \pm (1.96)(0.05)$
 $= 2.6 \pm 0.098 \Rightarrow (2.502, 2.698)$

$$99\% \Rightarrow Z_{\alpha/2} = 2.575 \Rightarrow 2.6 \pm (2.575)(0.05)$$

$$2.6 \pm 0.12875 \Rightarrow (2.47125, 2.72875)$$

- 9n CI, $\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$, $\boxed{e = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}}$

$$\boxed{\bar{X} \pm e}$$

→ margin of error

If $\bar{X}$ is used as an estimator for μ , we can be $100(1-\alpha)\%$ confident that the estimation error doesn't exceed $Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$$\boxed{|\bar{X} - \mu| \leq Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}}$$
 with confidence $(1-\alpha)$

- higher confidence $\Rightarrow$ larger $Z_{\alpha/2} \Rightarrow$ larger error margin
 large $\sigma \Rightarrow$ more variation $\Rightarrow$ more e
 large $n \Rightarrow \frac{1}{\sqrt{n}} \Rightarrow$ small e

- next, what should be our n st. estimation error does not exceed e .

$$Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq e \Rightarrow Z_{\alpha/2} \sigma \leq e \sqrt{n} \Rightarrow \frac{Z_{\alpha/2} \sigma}{e} \leq \sqrt{n}$$

$$\boxed{n \geq \left(\frac{Z_{\alpha/2} \sigma}{e} \right)^2}$$
 Round to next int.

eg CI = 95%, $e \leq 0.05$, $\alpha = 0.05$, $\frac{\alpha}{2} = 0.025$, $Z_{\alpha/2} = 1.96$

$$e = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \Rightarrow n = \left(\frac{Z_{\alpha/2} \sigma}{e} \right)^2$$

$$n = \left(\frac{1.96 \times 0.3}{0.05} \right)^2 = (11.76)^2 = 138.2916 \approx 139$$

- what if true σ is not known, $\boxed{\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}}$

$$Z_{0.025} = 1.96, \text{ but } t_{0.025, 6} = 2.447$$

t int. is wider than Z int. to account uncertainty in estimating σ .

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

$$P\left(-t_{\alpha/2, n-1} < T < t_{\alpha/2, n-1}\right) = 1 - \alpha$$

$$P\left(\bar{X} - t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

$$\therefore \left[\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} \right]$$

Ex. $n=7$ (9.8, 10.2, 10.4, 9.8, 10.0, 10.2, 9.6)

$df=6, \bar{X}=10$

$CI=95\%$

$\alpha=0.05, \alpha/2=0.025$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})^2$$

9.8 10 -0.2 0.04

10.2 10 0.2 0.04

10.4 10 0.4 0.16

$s^2 = 0.48 = 0.08$

9.8 10 -0.2 0.04

10.0 10 0 0

$S = 0.283$

10.2 10 0.2 0.04

9.6 10 -0.4 0.16

$10 \pm (2.447) \frac{0.283}{2.446} \Rightarrow$

10 ± 0.262

0.48

$\Rightarrow (9.738, 10.262)$

- We can have likewise one side bound as well

$L \Rightarrow \left(\bar{X} - t_{\alpha, n-1} \frac{S}{\sqrt{n}}, \infty \right)$

$U \Rightarrow \left(-\infty, \bar{X} + t_{\alpha, n-1} \frac{S}{\sqrt{n}} \right)$

note use use only α not $\alpha/2$

- if σ is unknown, but $n \geq 30$, with normal popula.

$\left(\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} \right) \rightarrow \left(\bar{X} \pm Z_{\alpha/2} \frac{S}{\sqrt{n}} \right)$

eg. $n = 500$, $\bar{x} = 501$, $s = 112$, $CI = 99\%$, $\alpha = 0.01$

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$\alpha/2 = 0.005$$

$$z_{0.005} = 2.575$$

$$501 \pm (2.575) \frac{112}{\sqrt{500}} \Rightarrow 501 \pm (2.575) \frac{112}{22.36}$$

$$501 \pm 12.898 \Rightarrow (488.102, 513.898)$$

- $SE(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}$ since $\text{var}(\bar{X}) = \frac{\sigma^2}{n} \Rightarrow SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$

- in general, estimate $\pm$ critical value $\times$ SE

for mean, known σ $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

for mean, σ unknown $\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$

- circularity problem in finding n with t

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \Rightarrow n = \left(t_{\alpha/2, n-1} \frac{s}{e} \right)^2$$

start with z estimate $\Rightarrow$ use n to get t critical value $\Rightarrow$ find n again.

- Width = $2 z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

confidence $\uparrow \rightarrow$ width $\uparrow$

$\sigma \uparrow \rightarrow$ width $\uparrow$

$n \uparrow \rightarrow$ width $\downarrow$